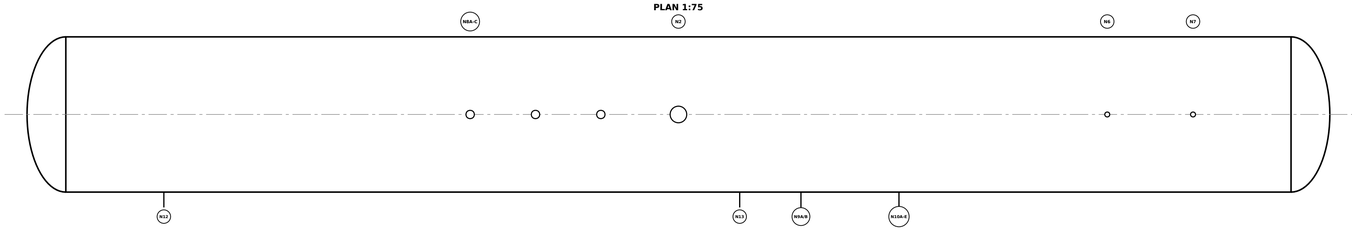
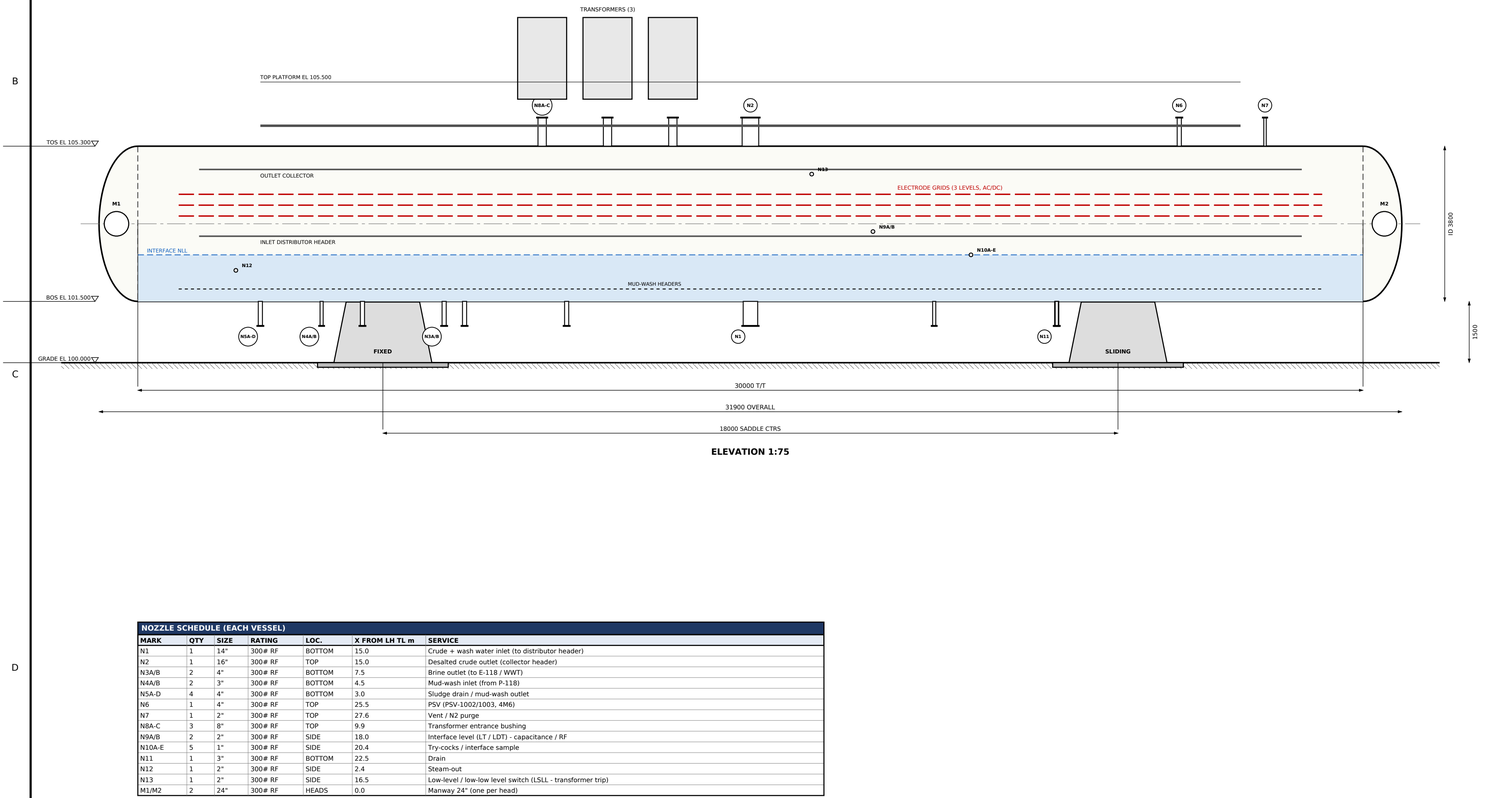
| DESIGN DATA (EACH) | |
|---|---|
| Service | **1st / 2nd stage electrostatic desalting** |
| Size | **ID 3800 x 30000 T/T** |
| Design P / T | **17.0 barg / 165 C** |
| Operating P / T | **11.5 barg / 136 C** |
| Material | **SA-516-70, CA 3.0 mm, E = 1.0** |
| Shell / head t | **28 / 28 mm (req. 27.0)** |
| Hydrotest | **22.10 barg (shop, horizontal)** |
| Residence time | **27.7 min** |
| Volume | **355 m3** |
| Insulation | **50 mm** |
| Weight empty / oper. / test | **137 / 414 / 487 t** |
| Code | **ASME VIII Div.1, U-stamp** |

**PLAN 1:75**




**ELEVATION 1:75**

**SECTION A-A 1:40 (TYPICAL)**




| NOZZLE SCHEDULE (EACH VESSEL) | | | | | | |
|---|---|---|---|---|---|---|
| **MARK** | **QTY** | **SIZE** | **RATING** | **LOC.** | **X FROM LH TL m** | **SERVICE** |
| N1 | 1 | 14" | 300# RF | BOTTOM | 15.0 | Crude + wash water inlet (to distributor header) |
| N2 | 1 | 16" | 300# RF | TOP | 15.0 | Desalted crude outlet (collector header) |
| N3A/B | 2 | 4" | 300# RF | BOTTOM | 7.5 | Brine outlet (to E-118 / WWT) |
| N4A/B | 2 | 3" | 300# RF | BOTTOM | 4.5 | Mud-wash inlet (from P-118) |
| N5A-D | 4 | 4" | 300# RF | BOTTOM | 3.0 | Sludge drain / mud-wash outlet |
| N6 | 1 | 4" | 300# RF | TOP | 25.5 | PSV (PSV-1002/1003, 4M6) |
| N7 | 1 | 2" | 300# RF | TOP | 27.6 | Vent / N2 purge |
| N8A-C | 3 | 8" | 300# RF | TOP | 9.9 | Transformer entrance bushing |
| N9A/B | 2 | 2" | 300# RF | SIDE | 18.0 | Interface level (LT / LDT) - capacitance / RF |
| N10A-E | 5 | 1" | 300# RF | SIDE | 20.4 | Try-cocks / interface sample |
| N11 | 1 | 3" | 300# RF | BOTTOM | 22.5 | Drain |
| N12 | 1 | 2" | 300# RF | SIDE | 2.4 | Steam-out |
| N13 | 1 | 2" | 300# RF | SIDE | 16.5 | Low-level / low-low level switch (LSLL - transformer trip) |
| M1/M2 | 2 | 24" | 300# RF | HEADS | 0.0 | Manway 24" (one per head) |

**NOTES**

1. Dimensions in mm, elevations in m; grade EL 100.000.
2. D-101A (1st stage) and D-101B (2nd stage) identical; series flow, wash water to 2nd stage, brine recycle to 1st stage.
3. Internals (distributor, electrode grids, collector, mud wash) by desalter licensor/vendor.
4. Transformers on top platform; vessel liquid-full - LSLL trips transformers.
5. Thicknesses per CFU-000-ME-CAL-001; datasheet CFU-000-ME-DS-002.
6. One saddle fixed, one sliding (slotted holes, PTFE slide plate).

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 2026-10-02 | Issued for review (FEED) |

GENERIC REFINERY - GULF COAST
**100 KBPSD CRUDE & VACUUM DISTILLATION UNIT**

# DESALTER GENERAL ARRANGEMENT

D-101A / D-101B ELECTROSTATIC DESALTERS (2-STAGE)

| DISCIPLINE | STAGE | SCALE | SHEET |
|---|---|---|---|
| MECHANICAL | FEED | 1:75 / 1:40 | 1 OF 1 |

| DRAWN | CHECKED | APPROVED | DATE |
|---|---|---|---|
| CLAUDE CODE | - | - | 2026-10-02 |

DRAWING NUMBER **CFU-100-ME-GA-007** REV **A**